JUBAID HOSSEN

Power Electronics & Embedded Systems Engineer

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Gazipur, Bangladesh

RESEARCH INTEREST

- Power Electronics Converters and Inverter Systems
- Digital Control of Power Electronic Systems
- Embedded Control for Power and Energy Applications
- Real-Time Embedded Systems for Power Electronics

ACADEMIC CREDENTIALS

Bachelor of Science in Electrical and Electronic Engineering Jan 2018 – Jan 2023
Jashore University of Science and Technology (JUST) Jashore, Khulna, Bangladesh
CGPA: 3.37/4.00(last 4 semester average 3.66)

WORK EXPERIENCE

Walton Digi-Tech Industries Ltd. (1.5 Year) Jul 2024 – Ongoing | Chondra, Gazipur
First Sr. Deputy Assistant Director
Electronics Research & Innovation(R&I)
Key Responsibilities:

- Hardware design for **power electronics converters, inverters**, and various home appliance products.
- Embedded systems architecture planning and real-time firmware development for power electronics applications.

RESEARCH EXPERIENCE

Industry Mentor, Power Electronics Research Group Sep 2025 – Present
Jashore University of Science and Technology (JUST)
Smart, Intelligent PV Inverter Research & Development
Supervisor: Md. Tarequzzaman , Lecturer
Department of Electrical and Electronic Engineering, JUST

Undergraduate Research Work Jan 2022 – Mar 2022
Jashore University of Science and Technology (JUST)
Thesis Title: "Tunable Properties of 2-D BiXS (X = Cl, I) Using First Principle Method for Photovoltaic Applications."
Thesis Supervisor: Dr. Md. Tanvir Hasan, Associate Professor
Department of Electrical and Electronic Engineering (JUST)

TECHNICAL SKILLS

Power Electronics & Hardware

- DC–DC converters, inverters, UPS systems
- Analog and digital circuit design
- MATLAB, LTspice, Proteus

PCB Design

- Multi-layer PCB design
- EMI/EMC-aware layout practices
- Altium Designer, EasyEDA

Embedded Systems & Microcontrollers

- MCU: STM32, ESP32, PIC MPU: NVIDIA Jetson Nano
- Frameworks: ESP-IDF, STM32-HAL, LVGL , The Yocto Project

Interfaces & Communication

- UART, SPI, I²C
- USB (CDC / custom class hardware support)
- ADC/DAC signal conditioning

DEVELOPED PRODUCTS

Uninterrupted Power Supply (UPS) Jul 2024 – Mar 2025
Model name: Walton arc | UX01, 800 VA/480W
Role: Firmware and Hardware Designer
Brief:

- Modified square wave, 4 MOSFET H-Bridge inverter design
- Bootstrap-type high-side MOSFET driver with integrated shoot-through protection
- DMA-based ADC sampling with precision timing
- Controlled by STM32F103C8T6 MCU.

Rechargeable Fan Aug 2024 – Feb 2025
Model name: *Walton W17OA-AS (17")*
Role: Hardware Designer
Brief:
- PWM-based speed control through MOSFET
- Ultra-low power consumption through careful component selection and deep sleep mode
- MCU and MOSFET-based BMS
- Controlled by Chinese APM32 MCU

PROJECTS

High Frequency DC-DC Converter Aug 2025 – Nov 2025
500W
Role: Hardware Developer
Brief: Collaborative research project on high-frequency DC–DC converters for inverter front-end design
Supervised by: Md. Tarequzzaman , Lecturer, Dept. of EEE, JUST

String Fault Detection System on PV Power Plant Sep 2024 – Dec 2024
Role: Hardware and Firmware Developer
Brief: The project was funded by the **Mongla 100 MW (Energon) Solar Power Plant** to detect PV faults on their solar arrays.
- Used ESP32 for wireless mesh network communications and fault detection.
- The system is connected to IoT via Thingspeak.

Supervised by: Dr. Engr. Emran Khan , Associate Professor and Chairman, Dept of EEE, JUST

Development of a Prototype-based Low-Budget Ventilator for COVID-19 Patients. Oct 2021 – Feb 2023
Role: Firmware Developer
Brief: This project was funded and supervised by Professor Md. Amzad Hossain, Dr. Engr. Postdoc to aid coronavirus-affected patients by providing a low-budget ventilator.
- Designed the servo-based airflow control mechanism.
- Ported firmware to **FreeRTOS**, improving real-time control performance and task scheduling.
- Participated in hospital testing and co-authored in publication (ICETSD-25).

VOLUNTEER EXPERIENCE

General Member
Shohayok Foundation, Community Welfare & Education Club
- Teaching basic primary education to underprivileged children
- Supporting financial assistance for patients requiring critical medical treatment.
- Organizing free medical camps and medicine distribution for low-income communities

General Member
JUST Robotics Society
- Gained hands-on experience in robotics, programming, and electronics.
- Contributed to team projects involving robot design, assembly, and testing.
- Supported the club in preparing for and participating in robotics competitions.
- Assisted in organizing workshops, seminars, and demonstrations.

REFERENCES

Md. Azmol Haque, *First Senior Additional Director, Research & Innovation (R&I)*, Walton Digi-Tech Industries Ltd.
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Dr. Engr. Imran Khan, *Associate Professor and Chairman, Department of EEE,* Jashore University of Science & Technology
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